Phase–Amplitude Coupling as a Mandatory, Gauge-Invariant Tokenization Operator for EEG

Anonymous authors
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Abstract

Neurophysiology gives EEG a strong structural prior: slow oscillatory phase gates fast oscillatory amplitude, a phenomenon known as phase–amplitude coupling (PAC). Yet architectures that inject this prior as a learnable module — an additive coupling bias, a gate, a PAC-weighted attention — consistently fail to separate from plain attention. We show why: in every such design the prior sits beside a free path that the optimiser can take instead, and it does. We therefore make the prior constitutive rather than modulatory. In our operator the token for a fast band is not a representation of that band which PAC then reweights; it is the product of that band's amplitude with the preferred-phase-aligned sum of all slower bands' phases. Above the slowest band no raw path exists, so the prior cannot be routed around. The construction has a further property no prior PAC model has: it is provably invariant to the arbitrary choice of phase reference, because the measured preferred phase cancels exactly the gauge rotation it induces on the learned phase features. On supervised EEG classification this operator improves Cohen's κ on TUEV from 0.4679 to 0.5493 at identical parameter count, and by a similar margin on sleep staging. Three parameter-matched controls each remove one ingredient and each falls back to baseline, isolating the measured preferred phase — not the multiplicative form, not the exposure of phase features, not coupling-strength weighting — as the source of the gain. [pending: pretraining and full foundation-model baselines]

1 Introduction

Foundation models for EEG (Yang et al., 2023; Jiang et al., 2024; Wang et al., 2025) tokenize the signal along time and space: a recording becomes a grid of (electrode, time-patch) tokens, and any frequency structure the model uses must be recovered by the encoder. This is a deliberate design choice — it makes the tokenizer domain-agnostic — but it discards a prior that neuroscience regards as one of the most robust organising principles of cortical dynamics.

That prior is phase–amplitude coupling (PAC): the amplitude envelope of a fast oscillation is modulated by the phase of a slower one (Canolty et al., 2006; Tort et al., 2010). PAC is directional (slow gates fast), it is measurable per electrode and per short time window, and it carries two distinct quantities — a coupling strength and a preferred phase, the phase of the slow rhythm at which the fast envelope peaks.

A natural idea is to build this into the architecture. Several works do (Lemkhenter & Favaro, 2020; Li et al., 2023b;a; Lee et al., 2026), and so did we, repeatedly. Our central negative finding is that a large class of these designs fails for a shared and predictable reason.

Priors with an escape route get optimised away. We ran four architectural injections of PAC on identical backbones and data (Section 3). An additive $\alpha \cdot \mathrm{PAC}$ bias drove its scalar $\alpha \to 0$. A sigmoid gate, initialised at its no-op value so it could "only help," never left that value across five datasets — an entire result table that turned out to measure plain attention. Two hyperparameter-free PAC-modulated attention operators, one soft and one a hard top-k topology, matched but did not beat plain attention, and did not beat their own label-destroying shuffle controls. In each case the module sat next to a path that could carry the same information without it, and training preferred that path.

Our design rule. If a prior can be bypassed, it will be. So we do not add PAC to a token; we define the token with it. For a band j above the slowest, the token is the elementwise product of that band's amplitude feature with a preferred-phase-aligned, coupling-weighted sum of the phase features of all slower bands. There is no raw waveform token beside it, no scale, no gate, no auxiliary branch. Every sample-dependent feature that reaches the encoder for $j > 0$ contains both factors, inseparably.

Gauge invariance. The phase origin of a band is arbitrary: change the reference electrode or the filter implementation and every phase shifts by a constant. A representation built on phase should not depend on that choice. Our operator satisfies this exactly and by construction (Section 5): a shift δ_i of band i's phase reference rotates the learned phase feature by $e^{i\delta_i}$ and rotates the measured coupling vector by the same $e^{i\delta_i}$, and the operator applies the conjugate of the latter to the former, so the two cancel. To our knowledge no prior PAC architecture has, or claims, this property; a survey of 28 related papers returns no occurrence of phase-reference invariance or equivariance in this sense.

Contributions.

1. A documented account of why architecturally injected PAC priors fail, from four controlled negative results, and the design rule this implies: the prior must be constitutive, not modulatory (Section 3).
2. A mandatory PAC tokenization operator, with a proof of exact invariance to the phase reference (Sections 4–5).
3. Evidence that isolates the responsible quantity. Three parameter-matched controls remove, respectively, the measured coefficients, the preferred-phase alignment, and the forced multiplication. All three fall back to baseline; only the complete operator gains (Section 6.3).
4. A capacity control showing the gain is not an artefact of model size: at 5.2× parameters the advantage is unchanged (Section 6.4).

2 Related work

EEG foundation models. BIOT (Yang et al., 2023) tokenizes per channel with an FFT and pretrains contrastively. LaBraM (Jiang et al., 2024) trains a vector-quantized tokenizer against the Fourier spectrum and then predicts masked codes. CBraMod (Wang et al., 2025) reconstructs masked raw patches with criss-cross spatial/temporal attention. All three tokenize time patches; none carries an explicit frequency axis, and none computes an interaction between bands as part of tokenization.

PAC in deep learning. PAC is not new to deep models, and we do not claim it is. PhaseSwap (Lemkhenter & Favaro, 2020) builds a pretext task from exchanging Fourier phase between segments. PACNet (Li et al., 2023b) uses event-related PAC to select subject-specific high-frequency bands ahead of an EEGNet-style decoder. Li et al. (2023a) assemble a complex-valued PAC image from Tort's modulation index and preferred phase and classify it with a complex CNN. SleepPACNet (Lee et al., 2026) is architecturally closest: it Hilbert transforms a low band, concatenates its real and imaginary parts with a high-band signal, and lets a CNN discover the interaction.

Our distinction from these is not the use of PAC but three structural properties: PAC is computed differentiably inside the network from a learnable filterbank rather than precomputed offline; the interaction is forced rather than made available; and the result is gauge-invariant. Notably SleepPACNet retains a separate raw-EEG branch alongside its PAC branch — by our analysis exactly the escape route that lets such priors be ignored — and its reported gain is 2.4 accuracy points with only one sleep stage significant after correction. We test the concatenative alternative directly inside our own architecture in Section 6.3.

3 Why architecturally injected PAC priors fail

[TODO: condense; currently the fullest version of the argument]

We summarise four attempts, all on the same tri-axial backbone, same data, same protocol, differing only in how PAC enters.

1. Additive bias. The frequency-axis mixer adds $\alpha \cdot C$ to the attention logits, C the measured coupling matrix, α learned. α decays monotonically toward zero during training.
2. Multiplicative gate. Attention probabilities are multiplied by $\sigma(w \cdot C)$ and renormalised, with w initialised at 0 so the module starts as an exact no-op and can "only switch gating on if it helps." Across five datasets and six layers the gate mean stayed at 0.5 to the fifth decimal — it never left initialisation. The recorded conclusions from those runs measured plain attention and were retracted.
3. Hyperparameter-free multiplicative modulation. Post-softmax attention weights are multiplied by row-mean-normalised coupling and renormalised. No learnable strength exists, so nothing can decay. It ties plain attention.
4. Hard topology. Coupling selects the top-k source bands per target; the rest are masked to $-\infty$. Also ties, and does not separate from a control in which the coupling matrix is shuffled.

The gate case is the most instructive. A module initialised at a no-op value has two indistinguishable outcomes: it correctly declined to act, or it never moved. Without an explicit check that the parameter changed, the design is untestable by construction. We recommend such a check as standard practice.

The common structure is that in (1)–(4) the encoder retains a complete, sample-dependent token for each band regardless of the PAC module, so the module is optional. The design rule that follows is the premise of this paper: to make a prior load-bearing, remove the alternative.

4 Method

4.1 Tri-axial analytic tokenization

Let $x \in \mathbb{R}^{B \times C \times T}$ be raw EEG. A learnable sinc filterbank (Ravanelli & Bengio, 2018), parameterised by band edges rather than kernel weights, produces n_b band-limited signals per electrode. A differentiable Hilbert transform yields, per band, a unit-modulus analytic phase $e^{i\varphi_i(t)}$ and an amplitude envelope $A_j(t)$. Time is divided into patches of length L; all quantities below are computed within one (electrode, patch) pair and we suppress those indices.

4.2 The measured coupling vector

For a slow band i and a fast band j we take the debiased mean vector (Canolty et al., 2006),

$$Z_{ij} \;=\; \frac{1}{L} \sum_t \big(A_j(t) - \bar{A}_j\big)\, e^{i\varphi_i(t)}, \tag{1}$$

retaining it as a complex number: $|Z_{ij}|$ is coupling strength and $\angle Z_{ij}$ is the preferred phase. Centring A_j removes the bias that a strictly positive weight would otherwise induce. Crucially we do not take the modulus at this point; discarding $\angle Z_{ij}$ is what left earlier operators with no phase geometry to steer by.

4.3 Two feature maps

A single bias-free convolution $\mathcal{L}$, shared between the real and imaginary parts, produces a complex phase feature; a separate convolution $\mathcal{A}$ produces an amplitude feature:

$$p_i \;=\; \mathcal{L}(\cos\varphi_i) + i\,\mathcal{L}(\sin\varphi_i), \qquad a_j \;=\; s \odot \mathcal{A}\big(\log(1 + A_j)\big). \tag{2}$$

The absence of a bias in $\mathcal{L}$ and its sharing across the two components are not incidental — they are exactly the conditions under which Section 5 holds. The diagonal s exists to match the parameter count of the raw-token baseline exactly.

4.4 The mandatory interaction

Only edges $i < j$ are admissible: slow phase drives fast amplitude. Coupling strengths are normalised per target band,

$$\alpha_{ij} \;=\; \frac{|Z_{ij}|}{\sum_{i'<j} |Z_{i'j}|}, \tag{3}$$

which makes the operator invariant to the overall scale of Z and therefore free of any temperature or strength hyperparameter. The token is

$$\boxed{h_j \;=\; \underbrace{a_j}_{\text{target amplitude}} \odot \sum_{i<j} \alpha_{ij} \underbrace{e^{-i\angle Z_{ij}}}_{\text{preferred-phase alignment}} \underbrace{p_i}_{\text{slower-band phase}}, \qquad h_0 \;=\; a_0 \odot p_0.} \tag{4}$$

The alignment factor is what makes the sum meaningful: different edges peak at different phases, and adding unaligned complex features would cause them to cancel. The slowest band has no slower driver and forms the root of the directed hierarchy. Tokens are mapped to $\mathbb{R}^{2K}$ by interleaving real and imaginary parts, then given a band embedding and an electrode-coordinate embedding, and passed to a factorised transformer attending along time, space, and frequency in turn. The frequency-axis mixer is plain attention: PAC enters the model in exactly one place.

4.5 No bypass

For $j > 0$ there is no raw token beside h_j, no learned coupling scale, no gate, no PAC branch and no auxiliary loss. The positional embeddings are constants and cannot carry sample-specific signal around the product. Because a_j and the aligned phase are combined multiplicatively, no downstream layer can recover either factor alone.

5 Exact invariance to the phase reference

Proposition 1. Let $\varphi_i \mapsto \varphi_i + \delta_i$ for arbitrary constants δ_i. If $\mathcal{L}$ is real-linear, bias-free, and shared between the real and imaginary components as in equation 2, then h_j is unchanged for all $j > 0$.

Proof. Expanding $\cos(\varphi + \delta)$ and $\sin(\varphi + \delta)$ and using linearity and the absence of a bias,

$$\mathcal{L}(\cos(\varphi_i{+}\delta_i)) + i\,\mathcal{L}(\sin(\varphi_i{+}\delta_i)) = e^{i\delta_i}\big(\mathcal{L}(\cos\varphi_i) + i\,\mathcal{L}(\sin\varphi_i)\big),$$

so $p_i \mapsto e^{i\delta_i} p_i$. From equation 1, $Z_{ij} \mapsto e^{i\delta_i} Z_{ij}$, hence $e^{-i\angle Z_{ij}} \mapsto e^{-i\delta_i} e^{-i\angle Z_{ij}}$, while α_{ij} is unchanged. The two rotations cancel in every term of equation 4. □

Numerically, applying independent random shifts to all bands changes $h_{j>0}$ by at most 8.3×10^{-7}, i.e. float32 rounding. The root token h_0 is gauge-covariant by definition — some band must fix the reference — and is excluded from the claim.

Two consequences are worth stating. First, this is a property of the operator, not something training must discover, which distinguishes it from a learned rotation of feature pairs that have no defined phase origin. Second, it constrains what a phase-based objective may target: a gauge-covariant quantity such as $\angle Z_{ij}$ itself cannot be predicted from a gauge-invariant representation, so only invariant targets are well posed.

Table 1: Tokenizer comparison. Each dataset is paired against a raw-token baseline retrained under the same protocol. TUEV and Sleep-EDF report κ, TUSZ reports PR-AUC. Parameter counts are identical within each pair.

Dataset	raw token	PAC interaction	Δ
TUEV (6-class, 16 ch, 200 Hz)	0.4679	0.5493	+0.0814
Sleep-EDF (5-class, 2 ch, 100 Hz)	0.5211	0.5732	+0.0521
TUSZ (binary, 9% positive)	0.5871	0.5690	−0.0181

Table 2: Parameter-matched controls on TUEV. Each removes exactly one ingredient of equation 4. All arms have 1,635,734 parameters except concat, which has 1.5% more.

Arm	What is removed	κ
raw baseline	the entire interaction	0.4679
uniform	measured α and phase alignment	0.4734
magnitude	phase alignment only	0.4826
concat	the forced product (replaced by a learned projection)	0.4838
scramble	the pairing of phases to edges (multiset preserved)	0.5099
measured	nothing	0.5493

6 Experiments

6.1 Setup

All comparisons are supervised training from scratch, identical data, splits, preprocessing, optimiser, schedule and seed; the only variable is the tokenizer. Selection follows Yang et al. (2023): Cohen's κ for multi-class, AUROC/PR-AUC for binary. All arms within a comparison have identical parameter counts (1,635,734 at the base capacity), so no result can be attributed to capacity. [pending: multi-seed; all numbers below are seed 0]

6.2 Main result

Table 1 shows gains on two datasets and a tie on the third. The Sleep-EDF cell is the most informative: sleep is the canonical PAC setting (delta phase modulating spindle amplitude, both inside the learned band range), it is the domain of the closest prior work, and it shares nothing with TUEV — different task, two channels rather than sixteen, 100 Hz rather than 200. The two-channel result also supports montage-agnosticism, since the electrode embedding is a function of coordinates rather than an index.

Under an identical pipeline, from-scratch BIOT reaches $\kappa = 0.4449$ and from-scratch CBraMod $\kappa = 0.5017$ on TUEV; our model reaches 0.5493 with 41% of CBraMod's parameters. We stress that published numbers for these models are not comparable to ours: the same released CBraMod checkpoint scores 0.6772 in its own pipeline and 0.4436 in ours, a gap larger than any effect we study, because its TUEV protocol runs at a different sampling rate. Only same-pipeline comparisons are reported here. [TODO: decide whether to also report our model inside their pipeline]

6.3 What is responsible for the gain

Table 2 is the core evidence. Three controls each fall back to the baseline, and each rules out a distinct alternative explanation.

uniform retains the full multiplicative topology and replaces only the measured coefficients by a uniform average. It ties the baseline (+0.0055). The multiplicative token structure

Table 3: Capacity control on TUEV. Arms are parameter-matched within each column.

	1.64M	8.6M	Δ (scale)
raw token	0.4679	0.4498	−0.0181
PAC interaction	0.5493	0.5379	−0.0114
difference	**+0.0814**	**+0.0881**	—

is therefore worth nothing on its own; what is worth something is the measured quantity placed inside it.

concat receives identical ingredients — the same a_j and the same gauge-invariant aligned phase — and differs only in combining them by a learned linear map instead of the forced product. It also ties the baseline, despite 1.5% more parameters. This is the SleepPACNet-style fusion evaluated inside our own architecture, and it isolates a single variable: whether the interaction is optional. It is the direct answer to the objection that our gain merely comes from exposing analytic phase.

magnitude keeps measured coupling strengths but drops the preferred phase, deterministically. Comparing the two deterministic controls decomposes the gain:

$$\underbrace{\text{magnitude} - \text{uniform} = +0.0092}_{\text{coupling strength}}, \qquad \underbrace{\text{measured} - \text{magnitude} = +0.0667}_{\text{preferred phase}}.$$

Almost all of the effect is attributable to the preferred phase — the most PAC-specific and least generic ingredient — rather than to weighting by coupling strength, which any reader would regard as the obvious move.

scramble preserves every $|Z_{ij}|$ and the entire multiset of preferred phases but permutes which edge owns which, redrawing the permutation every forward pass so the model cannot learn to invert a fixed one. It lands above magnitude, indicating that a stochastic complex rotation is itself mildly beneficial; this is why the deterministic magnitude arm, not scramble, is used for the decomposition above.

6.4 The gain is not a capacity artefact

A natural objection is that a small model may simply overfit less than a larger baseline. We scale both arms to 8,558,870 parameters (5.2×).

The advantage does not shrink; it is flat to marginally wider. Moreover, at a capacity exceeding CBraMod's, the raw tokenizer still loses to from-scratch CBraMod (0.4498 vs 0.5017) while the PAC tokenizer wins (0.5379). Building a larger model does not reproduce the effect; changing the token construction does.

6.5 Pretraining

[pending: This section is not yet run. The planned protocol pretrains on 18 corpora totalling ≈ 4,953 hours, selected for corpus diversity rather than volume, and evaluates against a supervised-from-scratch baseline at matched budget. Our present evidence is that on datasets where supervised training already works, single-corpus masked pretraining does not improve over no pretraining; whether multi-corpus pretraining changes this is the question the run is designed to answer.]

7 Limitations

Single seed throughout; the Sleep-EDF margin in particular is of the same order as its within-run validation variance. One TUEV cell was not measured. The learned band edges are unconstrained and could in principle cross, which would invalidate the slow-to-fast reading of $i < j$; final centre frequencies should be inspected before any physiological claim. A one-second patch contains few cycles of the slowest band, so the coupling estimate is noisy at that end. Finally, statistical PAC can arise from non-sinusoidal waveform shape or sharp transients (Kramer et al., 2008; Cole & Voytek, 2017; Aru et al., 2015); we therefore call the learned quantity a phase–amplitude dependency and make no claim that it identifies coupling between independent oscillators.

8 Conclusion

A prior that can be bypassed will be. Making phase–amplitude coupling constitutive of the token rather than a modifier of attention converts a repeatedly failing architectural idea into a working one, and the resulting operator is exactly invariant to the arbitrary phase reference. Parameter-matched controls locate the gain specifically in the measured preferred phase.

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